E-WASTE GOVERNANCE PLATFORM FOR INDIA

Final specification, compliance framework and national scale-up plan

Collect, verify, erase, pay, dispatch, recycle and certify: from the Thapar campus pilot to a multi-state operation

Prepared for	Energy and Environment (UEN008) project, Thapar Institute, Patiala
Date	20 October 2026
Built from	The team's governance model PDF, the team's workflow description, and official Indian sources (CPCB, MoEFCC, PIB, DoT, MeitY, CERT-In)
Audience	The engineer building the website, the faculty and Zero Waste Team approving it, and the auditor reviewing it
Status	Planning document. Not legal, tax or financial advice. Items marked CONFIRM need written confirmation before launch.

How to read this document

Facts carry a source number such as [S3]; the list is in section 17. Official government sources are listed first; commercial summaries are marked as secondary and should be checked against the gazette text.

Items marked [design] are my recommendations, not facts. Thresholds and capacity numbers are starting assumptions for the committee to change.

Money figures are left symbolic because real prices must come from local quotes.

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1. Executive summary

The platform follows every device from registration to certified recycling, with proof at each step. It has four working roles (user, collection engineer, factory engineer, admin or auditor) and one rule: a status can change only when its evidence exists, and every action is logged.

- **What changed since the first PDF.** The first document said India formally recycles about a third of its e-waste. That was the older picture. CPCB figures given to Parliament show 61.94% of generated e-waste collected, dismantled and recycled in FY 2023-24 and 70.71% in FY 2024-25 [S4]. The Ministry reports 9,79,080 tonnes recycled out of 14,14,645 tonnes generated in FY 2025-26 [S5]. So the national gap is no longer only collection volume. The gaps that remain are traceability, trusted data erasure, reuse, small-generator access and the accuracy of the modelled figures.
- **Why the model still makes sense.** CPCB estimates generation from producers' sales and average product life, so it is a modelled number, not a count of what households and offices discard [S4]. A platform that records real devices, real weights and real recycler receipts supplies the missing ground truth.
- **How to scale.** Hub-and-spoke: many collection points, a few city sanitisation centres, and CPCB-registered refurbishers and recyclers as the legal end of the chain. Start as a collection point working under agreement with registered partners, not as a recycler.
- **Biggest risks.** Legal status of the hub; personal-data duties under the DPDP Act (main duties enforceable from 14 May 2027 [S10]); payment fraud and informal-market competition; and unit economics.
- **Recommended path.** Campus pilot without cash (points or certificates), then city pilot with legal sign-off and cash payout, then state and multi-state expansion, each stage gated by measured results (section 12).

Five decisions needed before building starts

#	Decision	Who decides
1	Cash payout or points mode for the pilot	Faculty advisor, Zero Waste Team, legal advisor
2	Registered recycler and refurbisher partners, with their CPCB registration numbers	Committee
3	Whether the hub acts only as a collection point (recommended) or seeks its own registration	Legal advisor
4	Weight-mismatch tolerance, payment caps and the rate table	Committee, from the price survey
5	Hosting in an Indian cloud region and the data-retention periods	IT and legal

2. India context from official data

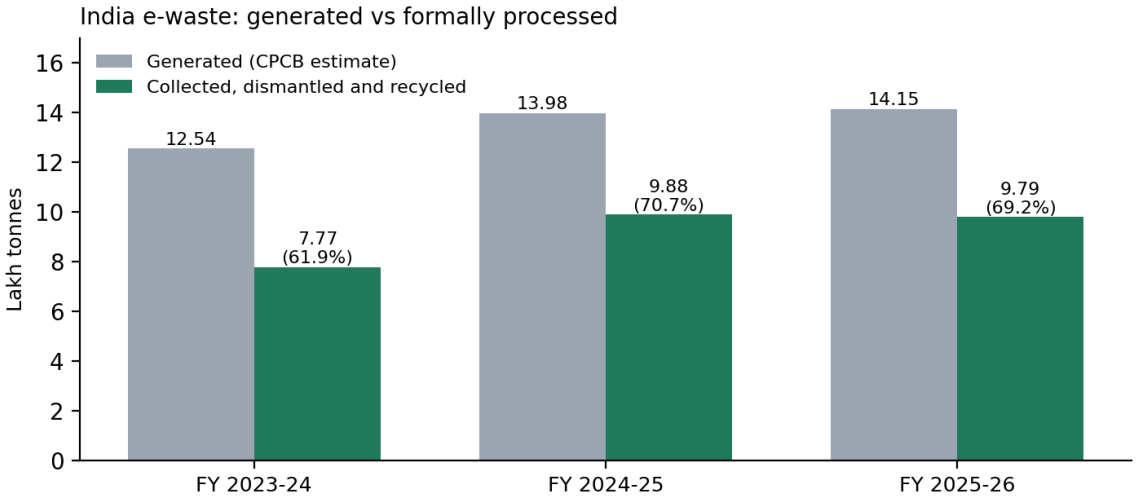


Figure 1. Generated e-waste is CPCB's estimate; processed is the quantity reported by registered producers, recyclers and dismantlers. Percentages computed from the quantities. FY 2023-24 processed quantity uses the CPCB percentage of 61.94 [S4]; FY 2024-25 and FY 2025-26 use the quantities in [S5] and [S6]. Small differences exist between releases.

Measure	Value	Source
E-waste generated, FY 2023-24	12,54,286 tonnes (CPCB estimate from producer sales and product life)	S4
E-waste generated, FY 2024-25	13,97,955 tonnes	S4
Share collected, dismantled and recycled or disposed	61.94% (FY 2023-24); 70.71% (FY 2024-25)	S4
FY 2025-26	14,14,645 tonnes generated; 9,79,080 tonnes recycled	S5
States leading recycling, FY 2025-26	Uttar Pradesh 3,76,896 t; Telangana 1,61,988 t; Haryana 1,35,951 t	S5
Solar PV modules recycled	About 1,064 t (FY 2023-24) and 1,057 t (FY 2024-25) by registered recyclers	S3
Under-collecting states relative to size	Maharashtra, Rajasthan and Punjab (analysis of CPCB data; secondary source)	S20

What this means for the design

- **Use official numbers as the denominator, not as proof.** The dashboard should show the platform's own collected and recycled weights beside the CPCB state figures, and label the CPCB figures as modelled estimates.
- **Plan transport lanes.** Registered recycling is concentrated in Uttar Pradesh, Telangana and Haryana [S5]. A Punjab hub will usually send material to Haryana or Uttar Pradesh, so dispatch records, geotags and weight reconciliation matter more over long distances.
- **Do not claim informal-sector numbers as fact.** Estimates of the informal share range widely across studies. The first PDF cites reports from over 50% to 90-95%. Treat these as unresolved and measure local prices and behaviour with your own survey.
- **Take no credit for volume already counted.** If a registered recycler already reports a device in its EPR figures, the platform should record it as traceability for that device, not as additional national recycling.

3. Legal and regulatory framework in India

The table lists what the platform must respect and what the software must therefore do. Legal text changes; the CONFIRM items need a lawyer or the regulator in writing.

Instrument and authority	Key points	Software and process consequence
E-Waste (Management) Rules, 2022, MoEFCC; in force 1 April 2023; amended in 2023 and March 2024; second amendment of 12 November 2024 set penalties under section 15 of the Environment (Protection) Act [S1, S7, S8]	Covers 106 equipment categories. Manufacturers, producers, refurbishers and recyclers must register on the CPCB portal and unregistered entities cannot lawfully operate [S2]. Producers meet targets through EPR certificates bought from registered recyclers; refurbishing certificates defer the obligation. Law-firm summaries state a storage limit of 180 days (extendable to 365) and collection targets of 60%, 70% and 80% of products placed on the market in the relevant earlier year, rising by FY 2027-28 [S9, S16] (CONFIRM against the gazette).	Store each partner's CPCB registration number and validity date; block dispatch to an unregistered or expired partner. Alert at 150 days of storage per device or batch. Keep weighbridge slips and recycler receipts as uploads.
CPCB EPR portal and SOPs [S2, S11]	Online registration for producers, recyclers, refurbishers and manufacturers. CPCB's FAQ says a refurbisher needs a valid consent to operate under pollution-control laws.	The platform does not issue EPR certificates. It reconciles its dispatch weights with partner records. Public portal API access is unconfirmed (CONFIRM); plan for CSV or manual reconciliation first.
Battery Waste Management Rules, 2022, MoEFCC and CPCB; notified August 2022 [S12, S13]	Cover all battery types, set EPR targets and require online registration; informal collectors and transporters may register on the portal.	Separate battery stream with its own status, sealed storage and a registered battery recycler. Option to onboard informal collectors as registered aggregators.
Digital Personal Data Protection Act, 2023 and DPDP Rules, 2025, MeitY [S10, S14]	Rules notified 14 November 2025. Phases: Data Protection Board provisions immediately; consent-manager registration from 14 November 2026; substantive duties on consent, notice, security and breach reporting from 14 May 2027. Penalties for failing reasonable security safeguards reach Rs 250 crore, and for failing to notify a breach Rs 200 crore, as set out in the first PDF (CONFIRM against the Act's schedule).	Build consent notice, purpose limits, retention, correction and erasure workflows from day one. Collect minimum personal data. Decide with a lawyer whether the operator is a data fiduciary or processor.
CERT-In Directions of 28 April 2022 under section 70B(6), IT Act [S15]	Specified cyber incidents must be reported to CERT-In within 6 hours of noticing them. ICT logs must be kept for a rolling 180 days within India.	Host in India or replicate logs in India; keep a written incident runbook with a named reporting officer; log retention of at least 180 days.
Sanchar Saathi, CEIR and Know Your Mobile, Department of Telecommunications [S17]	CEIR blocks lost or stolen phones; Know Your Mobile lets a person check whether an IMEI is genuine.	Record an IMEI check at intake for every phone. Blocked or doubtful IMEI forces HOLD. No public bulk API confirmed (CONFIRM with DoT).
Payments and tax	Payouts run on regulated rails (UPI, IMPS, NEFT) through a bank or an RBI-authorised payment service. GST and TDS treatment of payments for scrap or devices was not researched in detail.	Use a licensed payout provider. Have a chartered accountant confirm tax treatment before cash mode starts (CONFIRM).

A caution on 'E-Waste Rules 2026'

Some commercial websites describe new 'E-Waste (Management) Rules, 2026'. I could not confirm such a notification on official pages. The Ministry's rules page lists the 2022 Rules with the 2024 amendment, and a 2026 explainer from a compliance site also treats the 2022 Rules as operative [S8, S9]. Check moef.gov.in and the CPCB portal for the latest notification before launch.

4. Operating model for national scale

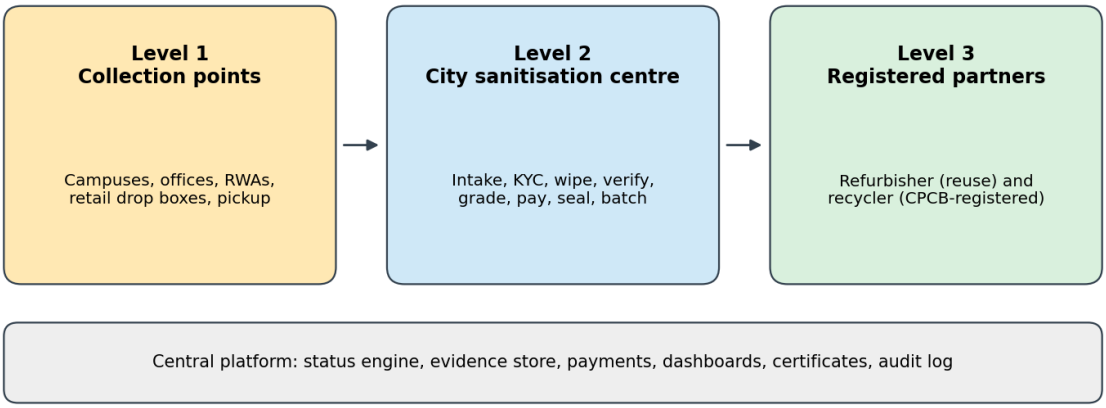


Figure 2. Three levels of physical operation under one central platform.

Legal positioning options

Option	What it means	Assessment
A. Collection point under agreement with a registered recycler or refurbisher	The hub receives devices, runs intake, wiping and evidence, and hands over to the registered partner under written agreement.	Recommended for the pilot and city phase. Lowest registration burden. CONFIRM that the Rules and the partner's authorisation allow this arrangement.
B. Register as a refurbisher on the CPCB portal	Needs consent to operate under pollution-control laws, CPCB registration, annual maintenance charges and its own compliance returns [S11].	Consider once reuse volume is real and a refurbishing workshop exists.
C. Become a recycler	Plant, pollution control, authorised capacity, recovery targets.	Not recommended for this project. Capital heavy; existing registered recyclers already exceed collection.

Levels and responsibilities

Level	Examples	Does	Does not
1 Collection points	Campus bins, office drop boxes, residents' associations, retail partners, scheduled doorstep pickup	Booking, sealed intake, ownership declaration, photo, tamper-evident bag	Open, wipe or dismantle devices
2 City sanitisation centre	Staffed room with secure cage, wiping stations, scales, camera capture	Registration against the user's phone, IMEI check, wipe, verification video, payment, dispatch evidence, batching	Recycle or resell without a registered partner
3 Registered partners	CPCB-registered recycler; registered refurbisher; registered battery recycler	Authenticity and weight check, refurbishment, recycling, recovery records, EPR documentation	Accept devices without a matching record

Livelihoods and informal collectors

Formalisation can displace informal collectors. The Battery Rules already allow informal collectors and transporters to register on the portal [S12]. Design the platform so a registered aggregator account can bring devices to a hub and be paid on the same terms, with the same evidence rules. Thapar's inclusion of its own waste staff is the model cited in the first PDF.

5. End-to-end workflow and statuses

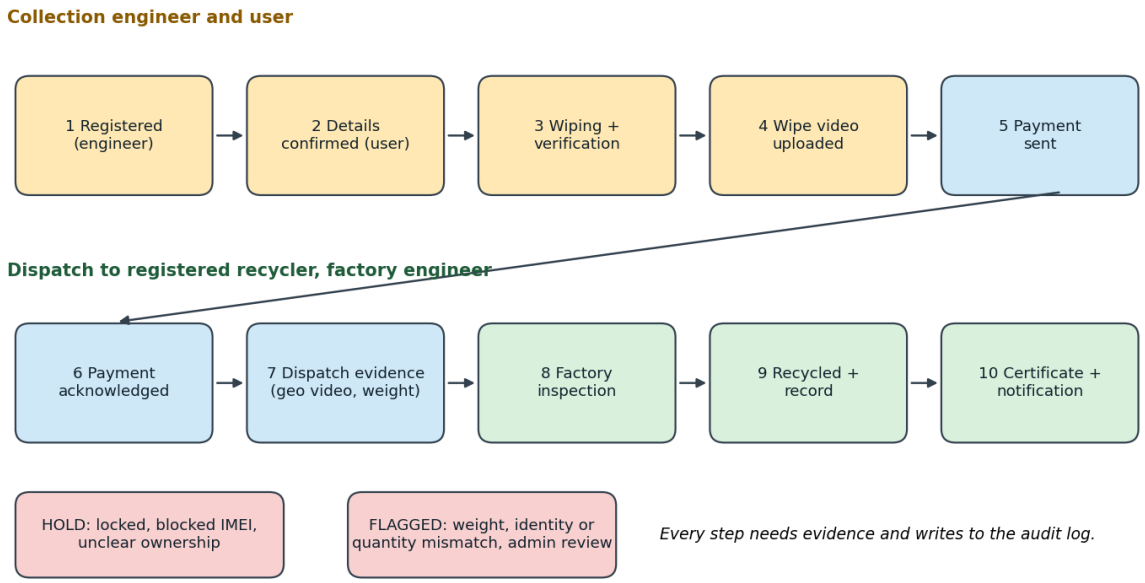


Figure 3. Ten forward stages and two side branches. Amber: collection engineer and user. Blue: payment and dispatch. Green: factory side. Red: exceptions.

Status	Set by	Evidence required to enter
REGISTERED	Collection engineer	Device details entered against the user's phone number; intake photo; signed ownership declaration; IMEI check result for phones
DETAILS_CONFIRMED	User	OTP or tap confirming the details shown
HOLD (branch)	Engineer or admin	Reason code: locked and password forgotten, blocked IMEI, unclear ownership. Payment disabled. Owner informed; police referral if required
WIPING	Collection engineer	Method from the dropdown for the storage type, tool name and version, start time
WIPE_VERIFIED	Collection engineer	Verification result, sampled recovery test, wipe video uploaded. User is notified
PAYMENT_SENT	Collection engineer	Amount, method chosen by the user, transaction reference
PAYMENT_ACKNOWLEDGED	User	User confirms receipt in the app
READY_FOR_DISPATCH	Collection engineer	Live geotagged video of the device on the scale, photo, weight and quantity entered. Added to an outgoing batch
FACTORY_INSPECTION	Factory engineer	Batch received; authenticity, weight and quantity checks
FLAGGED (branch)	Factory engineer or system	Reason code and note; goes to the admin review queue; recycling blocked
ACCEPTED	Factory engineer	All green flags
RECYCLED	Factory engineer	Input weight and output weights by fraction; batch reference
CERTIFIED	System	Certificate generated; user notified

Order rules. Payment is blocked until WIPE_VERIFIED. Dispatch evidence is blocked until PAYMENT_ACKNOWLEDGED. If the user does not acknowledge, send reminders at 3 and 7 days, then let the admin resolve it with a note. A device cannot skip a status, and the server rejects any jump.

6. Functional specification by role

6.1 Admin and auditor home page (restricted)

This is the only landing page for the Master Admin and the Auditor, and the server returns 403 to everyone else. The auditor is read-only and independent of the operators, as the governance model requires.

Area	Contents
Summary cards	Devices registered, kg collected, devices wiped, amount paid, kg recycled, recycling rate (recycled kg divided by collected kg), flagged devices, open holds, average days per stage
Graphs	Collected versus recycled by month; funnel by status; devices by category; recovered material by fraction; flag rate by reason and by engineer; payments by method and month; dispatch map from geotags; optional energy and carbon panel
Tables	Users with their devices, status, payment and certificate; collected goods; recycled goods; flag review queue; audit log. Phone and email are masked by default; each unmask is logged
Drill-down	National, state, city, hub, engineer, device. Filters by date, category, status, partner
Exports	CSV and PDF with current filters, for audit and for reconciliation with recycler records

6.2 User portal

- Login with registered mobile number or email by one-time password. No stored password.
- Each device card shows brand, model number, UEFI firmware model and version, and condition: working; locked with password forgotten; not booting or opening at all. The engineer reads these from the device and enters them, because a website cannot read them remotely. The user only confirms them.
- Stage tracker with dates, a notification when the device is cleaned (with the wipe video), the payment preference form (UPI ID or bank account; editable until PAYMENT_SENT), an acknowledge-payment button and the certificate download.
- Languages: English and Hindi at launch, then Punjabi and other regional languages with the state rollout [design].

6.3 Collection engineer console

- Find or create the user by phone number and verify with OTP so the right person is linked.
- Register the device: category, brand, model, serial, IMEI, UEFI model and version, storage type and size, battery present, condition, intake photos, ownership declaration.
- Wipe: method from a dropdown tied to storage type (section 9), tool and version, times, kWh if measured; verification result and sampled recovery test; wipe video upload.
- Payment: suggested amount from the rate table; the engineer confirms, pays on the user's chosen method and records the reference.
- Dispatch evidence icon: appears only after PAYMENT_ACKNOWLEDGED; opens the live capture screen; then 'Ready for dispatch' joins the batch list.

6.4 Factory engineer console

- Incoming batch list with collection-side data and evidence.
- Authenticity check (serial or IMEI, brand, model must match). Weight and quantity check with the difference shown.
- Automatic flag when the weight differs by more than the threshold, or identity or quantity does not match. Suggested starting rule: more than 10% or 100 g, whichever is larger, adjustable by the Master Admin [design].
- Green-flag checklist, recycling record by fraction, then 'Mark recycled', which issues the certificate and notifies the user.

7. Evidence capture, certificate and notifications

Wipe video

The video shows the device identifier, the wipe tool's start screen with the method name, and the final verification result. It must not show personal files, accounts or messages. Show an on-screen checklist at upload, preview before saving, reject files that are too short or too large, and store a SHA-256 hash. A video proves a process ran; it does not prove nothing remains. That is why the sampled recovery test and independent audit stay as separate recorded steps.

Dispatch evidence (the geo icon)

- Record live in the browser: no gallery upload, so old files cannot be reused. Needs HTTPS.
- Video of the device on a scale with the display readable, plus quantity; one photo of the device.
- Capture GPS latitude, longitude and accuracy plus a server timestamp. Reject if location is off or accuracy is worse than a set limit (suggested 100 m) [design].
- Show the location on a map in the device record and on the dashboard map.
- Weights are typed in and also visible in the video, so the factory can compare the three.

Certificate

PDF containing certificate ID, user name, device brand and model, partly masked serial or IMEI, date of data erasure and method, recycler name and CPCB registration number, recycling date and batch reference, and a statement that the device was processed through a registered recycler. A QR code opens a public verification page that shows only the certificate ID, device model, status and dates, with no personal details. Sign the PDF or store its hash so it can be checked later.

Notifications

SMS (using templates registered as the telecom rules require), email and in-app messages at: device registered, details awaiting confirmation, wipe verified, payment sent, dispatched, flagged (to the admin), recycled, certificate ready. Add WhatsApp later if a business-approved channel is available [design].

Storage and retention

Private object storage in an Indian region; short-lived signed links only; hashes at upload; retention schedule set by the committee (suggested: keep evidence for the audit period, then delete) [design]. Keep ICT logs for at least 180 days within India [S15].

8. Payments for India

The first PDF shows that a small payment alone will not beat informal buyers, who are reported to pay at least double what formal recyclers offer. Payment therefore has to be paired with a trusted wipe, convenience, a certificate and fair reuse value.

Pricing rule

Offer to the contributor = max(S, a x R) minus H, with a floor and a cap. S is scrap value (weight times the recycler's quoted rate for that category). R is expected resale value after grading. a is the share of R passed to the contributor. H is handling cost per item. Keep the rate table (category by condition: working, locked, not booting) editable by the Master Admin and fill it from at least ten local scrap dealers and two refurbishers.

Payout design

Item	Specification
Methods	UPI ID or bank account, chosen by the user in the portal. Points mode for the campus pilot.
Rails and provider	A bank or RBI-authorised payment service with a payout API. For the pilot, manual UPI transfer with the reference typed in is acceptable if the committee approves.
Verification	Name-match or small test credit to confirm the payout account belongs to the user. Do not collect or store Aadhaar numbers unless a law requires it [design].
Limits	Per-person and per-day caps; manual approval above the cap; no cash for items of unclear origin.
Records	Transaction reference, receipt visible to the user, reconciliation file matched to bank statements monthly.
Fraud checks	IMEI and ownership declaration; duplicate serial or IMEI detection; repeated submissions from one account; photos at intake; refuse unclear items.
Grievance	In-app dispute button; response target of 7 days [design]; escalation to the committee.
Tax	CONFIRM GST and TDS treatment with a chartered accountant before cash mode.

Funding sources

Source	Reliability	Note
Recycler payment for material by weight and content	Variable with metal prices	Basic income stream
Resale or donation of graded A and B devices through a registered refurbisher	Medium	Pay the contributor more for reuse than for scrap; this also discourages churn
Paid secure disposal for companies and institutions (IT asset disposition)	Strong	Companies value the wipe certificate and chain of custody
EPR-linked income from producers or recyclers	Uncertain	A 2026 field study cited in the first PDF argues certificates reward paperwork more than collection
Institutional budget and CSR	Good for pilots	Use for the first two phases

9. Data erasure standards and methods

The wipe dropdown follows NIST SP 800-88 Rev. 2 (September 2025) for policy and IEEE 2883-2022 for technique, as summarised in the first PDF. Read the standards themselves before quoting them; the first PDF's sources for them were vendor and commentary sites.

Media	Method	Verification
Hard disk	Overwrite acceptable for low sensitivity (Clear); purge or destroy for sensitive data	Read-back sampling
SATA SSD	ATA Secure Erase or cryptographic erase (Purge); plain overwrite is not enough	Tool status plus sampled recovery test
NVMe SSD	NVMe Sanitize (block erase or crypto erase); a format only clears the namespace	Controller completion plus sampled test
Phone or tablet	Remove accounts and activation lock, encrypt, factory reset; vendor cryptographic erase where available; destroy if it cannot be reset	Boots to setup screen; sample forensic check
SD card, USB stick	Treat as flash: purge by the device method or destroy	Sample check
Network gear, printers	Reset to defaults, then sanitise internal storage	Configuration check
Dead or failed media	Destroy the storage part only; recycle the rest	Photo and weight log

A factory reset is only the lowest level (Clear). The software warns when it is the only method used on a flash device. A locked device whose password is forgotten cannot be wiped and goes to HOLD; the owner is asked to recover access, or the storage is destroyed with the owner's written consent.

Wipe sampling at scale [design]

Run a recovery test on a random sample of wiped devices, for example 5% in the pilot, 2% at city scale, and 100% for any device flagged high sensitivity or for any engineer whose first-month fail rate is above the committee's limit. The auditor performs a further independent sample. Any failed sample sends that day's batch back for re-checking.

10. Technical architecture for scale

Layer	Choice	Reason
Front end	React or Next.js; Tailwind; charts with Recharts or Chart.js; maps with Leaflet and OpenStreetMap	Common skills; no licence cost; works on phones
Back end	Node.js (NestJS) or Python (Django); REST API with role checks on every route	Mature, auditable, fits a student team
Database	PostgreSQL with row-level access rules and a read replica for dashboards	Relational, audit-friendly; scales to the volumes below
Files	S3-compatible private bucket in an Indian region; lifecycle rules; signed links	Video-heavy; India hosting supports log and data location duties
Jobs and queues	A queue for video processing, notifications, certificate PDFs and exports	Keeps the interface responsive at peak
Identity	OTP for users; password plus second factor for staff; single role per account	No password database for the public
Observability	Central logs retained 180 days or more in India; alerts; incident runbook	CERT-In duties [S15]
Environments	Development, staging and production; daily backups; restore drill every quarter	Recoverability

Capacity planning [design assumptions]

These numbers are planning assumptions to size storage and cost, not forecasts.

Stage	Devices per year	Average per day	Evidence storage per year
Campus pilot	1,000	About 3	About 60 GB
City phase	20,000	About 55	About 1.2 TB
State phase	100,000	About 275	About 6 TB
Multi-state	500,000	About 1,400	About 30 TB

Basis: two videos and two photos per device at roughly 60 MB in total after compression. Compress on upload, keep originals for the retention period only, and design for peaks of about five times the daily average.

Integrations roadmap

Integration	Phase	Status
SMS provider and email service	Pilot	Available commercially
Payout provider (UPI, IMPS)	City phase	Choose a regulated provider; contract and approvals needed
Sanchar Saathi IMEI check	Pilot (manual)	No public bulk API confirmed; request integration from DoT later
CPCB EPR portal reconciliation	City phase (CSV)	API access unconfirmed
Recycler weighbridge or ERP feeds	State phase	Per partner agreement
Single sign-on for corporate clients	Multi-state	Optional

11. Security, privacy and audit controls

Risk or duty	Control built into the platform
Data exposed in transit or storage before wiping	Owner pre-wipe instructions; tamper-evident bags; secured cage; logged chain of custody; encryption where the device supports it
Personal data held by the platform	Minimum fields; role-based masking; encryption at rest and in transit; payout details stored separately and encrypted; retention schedule; deletion on request where the law allows
Consent and notice (DPDP)	Plain-language notice in the user's language at registration; recorded consent; purpose limits; correction and erasure requests handled in the portal; response within the period the Rules set
Breach response	Runbook; named officer; 6-hour CERT-In reporting for specified incidents [S15]; notification of affected users as the DPDP Rules require from 14 May 2027 [S10]
Certificate privacy	Public verification page shows no name, phone or email
Insider misuse	Role-based access enforced on the server; every view of evidence, unmask, export and edit logged; two-person approval for payment above cap and for flag resolution
Tampering with evidence	SHA-256 hash at upload; live capture only for dispatch evidence; server timestamps
Single point of failure	Backups, restore drills, manual fallback forms for intake and dispatch
Physical safety	Training; PPE; fire-safe battery storage; no dismantling at hubs. The first PDF cites 245 lithium-battery fires at 64 waste facilities in a US EPA study

12. Rollout roadmap with go and no-go gates

Phase	Scope	Payment mode	Gate to move on (suggested thresholds [design])
0. Campus pilot (this semester, about 13 weeks)	Thapar campus: bins, booking, passport register, intake, wipe demo with consenting owners, hold-and-escort of other devices to the registered recycler, awareness campaign, baseline and end surveys	Points or certificates; no cash	Written approvals; zero data incidents; all dispatches reconciled to recycler receipts; price survey complete; survey shows higher trust in the process
1. City pilot (months 5 to 12)	Patiala: institutions, offices and a few residents' associations; one registered recycler and one registered refurbisher under agreement; payout live	Cash after legal and tax sign-off	Weight-mismatch flags under 5%; wipe sample failures under 1%; payout disputes under 1%; cost per device known; audit clean
2. State phase (year 2)	Punjab and the Chandigarh region; two or three city centres; corporate IT asset disposition contracts	Cash and points	Positive or break-even unit economics on the B2B mix; second recycler partner; independent audit report published
3. Regional (years 3 to 4)	Punjab, Haryana, Delhi region, Uttar Pradesh: where most registered recycling capacity sits [S5]	Cash and points	Platform handles about 100,000 devices a year with the response targets met; incident drill passed; DPDP duties fully in force and compliant
4. National (year 5 onward)	Top generating cities and states; franchise or partner-run hubs on the same platform	As approved	Reconciled data shared with regulators if requested; governance model replicated by each region

Pilot semester plan (from the first PDF)

Weeks	Activities	Output
1-2	Permissions, safety rules, scope; draft model	Approved scope
3-4	Audit and survey; price-benchmark survey; select registered partners	Baseline data and quotes
5-7	Install bins; booking form and QR register; hostel champions; awareness launch	Pilot live
8-10	Intake and triage; repair cafe; wipe demo; computer-vision classifier training and testing	Demo-ready prototypes
11-12	Second survey; weigh outputs; energy and carbon sheet; reconcile with recycler receipts	Results
13	Record video; slides; rehearse viva	Submission

13. Dashboard indicators and anomaly rules

Indicator	Definition	Why it matters
Collection volume	Kg and devices received per hub, week and category	Core output
Recycling rate (platform)	Kg recycled divided by kg collected, from recycler records	Mirror of the national percentage [S4]
Wipe pass rate	Sampled recovery tests passed divided by tests run	Trust
Flag rate	Flagged devices divided by devices inspected, by reason and engineer	Fraud and quality
Turnaround	Days between status changes, median and 90th percentile	Service quality
Reuse rate	Devices graded A or B and reused divided by devices received	Environmental benefit
Battery share	Kg and count of batteries handled in the sealed stream	Safety planning
Payout per kg and per device	Money paid divided by weight and by count	Economics
Cost per device	Total operating cost divided by devices	Viability
Net emissions benefit	Avoided emissions from reuse minus wipe energy minus transport (method in the first PDF, with a stated displacement factor)	Energy and Environment link

Anomaly rules [design]

- Same IMEI or serial registered twice, or by two users.
- Dispatch geotag outside the registered hub's area, or the same coordinates for many unrelated devices.
- An engineer whose flag rate, weight differences or wipe-sample failures are well above the group's.
- A user with repeated submissions near the daily cap.
- Weights that are identical across many devices of different categories.
- Batches held longer than 150 days (storage limit alert).

Energy and environment calculation

- Avoided emissions = devices reused x emissions of making a comparable new device x displacement factor. The first PDF gives about 227 to 270 kg CO2 for manufacturing a laptop (ASU study) and a conservative displacement factor of 0.5; state any factor you use.
- Hub energy = measured kWh per wipe x devices wiped x grid emission factor; measure with a plug-in meter.
- Transport emissions = kilometres x vehicle emission factor; consolidate trips.
- Material recovery = kg handed to registered recyclers x the recovery fractions in the recycler's report. Do not claim more than the recycler documents.

14. Governance and organisation

Body or role	Responsibility
Campus or regional E-Waste Committee	Sets prices, accepted items, safety limits, data rules, partner selection; reviews the audit; publishes a scorecard. Includes faculty, Zero Waste Team, IT, legal advice and student seats
Master Admin	Creates staff accounts; edits thresholds, rate table, partner list; resolves flags with a written note
Independent auditor	Read-only access; samples wiped devices, checks logs, reconciles weights and payments; reports to the committee, not to the operators
Hub operator (student or staff team)	Runs intake, wiping and payouts under faculty supervision; trained volunteers only; no dismantling
Compliance officer (from the city phase)	Tracks partner registrations and validity, DPDP duties, CERT-In incidents and tax
Support desk (from the city phase)	Handles disputes and user questions within the response target
Registered partners	Refurbisher and recycler; issue receipts, weighbridge slips and recovery reports

Scorecard. Publish monthly: devices received, kg to recycler, reuse count, wipe pass rate, flag rate, average payout, number of disputes. Do not publish personal data.

15. Risk register

Risk	Effect	Mitigation
Hub's legal status unclear	Penalty or forced closure	Operate as a collection point under agreement; legal opinion in writing; CPCB registration check for every partner
Data breach before or after wiping	Penalties and loss of trust	Secure cage, sealed bags, encryption, minimal data, breach runbook, Indian hosting
Cannot out-pay informal buyers	Low-value waste only	Compete on certificate, convenience, reuse value; target institutions and companies
Stolen or counterfeit devices bought	Fraud, police issues	IMEI check, ownership declaration, HOLD branch, caps, refuse unclear items
Fake or inflated weights	Payment and reporting errors	Live scale video, geotag, factory comparison, automatic flags
Wipe imperfect on SSDs and phones	Residual data	Standards-based methods, sampled recovery tests, destroy storage when in doubt
Funding gap	Hub cannot continue	B2B certified disposal, resale margin, CSR, recycler payments; pilot on budget
Battery fire	Injury and property loss	Battery check at triage, sealed fire-safe storage, trained staff
Partner loses registration	Chain of custody breaks	Validity dates in the system; block dispatch automatically
Student turnover	Knowledge loss	Standing club, written procedures, handover checklist
Platform outage or attack	Operations stop	Backups, restore drills, paper fallback, incident drill
Over-reliance on modelled national data	Wrong claims	Label CPCB figures as estimates; report platform data separately

16. Procedure for the engineer building the website

Build in this order. A step is done only when the 'done when' test passes.

Step	Work	Done when
1	Database, roles, OTP login for users, password plus second factor for staff, audit log	Each role logs in and sees only its pages; dashboard URL returns 403 to others
2	Device registration against phone number; user confirmation; IMEI check field	Engineer registers; user sees and confirms
3	Status engine and stage tracker	Invalid status jumps are rejected by the server
4	Wipe records, evidence upload with hashing, notifications	User is informed at WIPE_VERIFIED
5	Payment preference, payment record, acknowledgement; points and cash switch	Payment is blocked until wipe verified
6	Live geotagged capture and dispatch batches	Evidence unlocks only after acknowledgement; location off blocks saving
7	Factory checks, automatic flags, recycling record by fraction	Flag fires on threshold breach
8	Certificate PDF, QR and public verification page	Verification page shows no personal data
9	Admin dashboard, tables, exports, anomaly rules	Dashboard totals equal sums of the raw records
10	Security review, restore drill, load test at five times the planned daily volume, pilot run	All acceptance tests below pass
11	Scale work: Indian-region hosting, read replica, queue workers, regional languages, partner feeds	Targets for the current phase met in staging

Acceptance tests

- A user cannot see another user's device or open the dashboard.
- Payment cannot be sent before wipe verification; dispatch evidence cannot be saved before payment acknowledgement.
- A blocked or doubtful IMEI forces HOLD and disables payment.
- A weight difference above the threshold creates a flag automatically; recycling is blocked while flagged.
- Location off or poor accuracy blocks dispatch evidence.
- The certificate verification page shows no name, phone or email.
- A partner with an expired registration cannot be chosen for dispatch.
- Every status change, unmask and export appears in the audit log.
- Backups restore in a drill and logs are kept for at least 180 days.

Data model (main tables)

users, payment_preferences (encrypted), devices, ownership_declarations, wipe_records, payments, dispatch_records, batches, partners (with CPCB registration number and validity), factory_checks, flags, recycling_records, certificates, files (with SHA-256), notifications, settings (threshold, rate table, caps, retention, payment mode), audit_log. Field lists are in the earlier build specification and carry over unchanged, with the addition of partners, batches and hub identifiers.

17. Items to confirm in writing, and sources

Confirm before launch

#	Item	Ask
1	Whether a hub that is not itself registered may collect devices, wipe them and hand them to a registered recycler or refurbisher	Lawyer, CPCB or the State Pollution Control Board
2	Current storage limit, targets and any 2026 changes to the E-Waste Rules	MoEFCC and CPCB notifications
3	Whether the operator is a data fiduciary or processor, and consent wording	Lawyer
4	GST and TDS on payments and on resale	Chartered accountant
5	Whether the CPCB portal and Sanchar Saathi offer any API or bulk access	CPCB and DoT
6	Institutional approvals, insurance and fire-safety rules for holding batteries	University administration
7	Written agreement and registration numbers from the recycler and refurbisher	Partners

Sources

Official and government sources are listed first. Secondary sources (law-firm and compliance-site summaries) were used only where the official text was not retrieved; confirm those points against the gazette.

ID	Source	Used for
S1	MoEFCC, Rules and Regulations page, moef.gov.in/rules-regulations-3 (lists E-Waste (Management) Amendment Rules, 2024)	Rules in force and amendments
S2	CPCB, FAQ on E-Waste (Management) Rules, 2022, cpcb.gov.in/uploads/Projects/E-Waste/FAQ_ewaste_23012024.pdf and eprewaste.cpcb.gov.in/assets/PDF/faqewaste.pdf	Who must register; refurbisher consent to operate
S3	CPCB report to the National Green Tribunal, EA 04 of 2024 in OA 512 of 2018, greentribunal.gov.in	Generation, recycling and solar PV figures
S4	PIB, Ministry of Environment, answers in Parliament, PRID 2147876 and PRID 2155124, pib.gov.in	12,54,286 t and 13,97,955 t generated; 61.94% and 70.71% processed
S5	Ministry data reported by ETV Bharat, March 2026 (FY 2025-26 figures and state split)	14,14,645 t generated; 9,79,080 t recycled; leading states
S6	As S5; FY 2024-25 processed quantity 9,88,479 t from CPCB data compiled by Dataful (secondary)	Chart value for FY 2024-25
S7	MoEFCC notification G.S.R. 699(E) of 12 November 2024, via TeamLease RegTech summary (secondary)	Second Amendment, penalties under section 15
S8	ChemLinked regulatory database entries for the 2022 Rules and amendments (secondary)	Dates of 2022 Rules and 2023, 2024 amendments
S9	Saikrishna and Associates (2022) and Global Risk Community (2026) summaries of the 2022 Rules (secondary)	Targets 60, 70, 80%; 2022 Rules still the operative framework
S10	MeitY, DPDP Rules, 2025, notified 14 November 2025; PIB release 17 November 2025; phase dates per AMS Shardul (secondary)	DPDP timeline: 14 Nov 2025, 14 Nov 2026, 14 May 2027
S11	CPCB EPR E-Waste portal SOPs for refurbisher, recycler and producer, hrocmms.nic.in and aboutepr.com (secondary)	Registration process
S12	Battery Waste Management Rules, 2022 (MoEFCC), via Down To Earth and IEA policy database (secondary)	Battery EPR and registration; informal collectors
S13	CPCB battery EPR portal notice via Lexplosion (secondary)	Battery registration process
S14	Anantam IAS and S.S. Rana summaries of the DPDP Rules (secondary)	Phased commencement details
S15	CERT-In Directions of 28 April 2022, cert-in.org.in; summaries by Mondaq and AZB (secondary)	6-hour reporting; 180-day logs
S16	GreenSutra EPR explainer, 13 June 2026 (secondary)	106 categories; refurbishing certificates
S17	Department of Telecommunications press release, 16 May 2023, sancharsaathi.gov.in	CEIR and Know Your Mobile

ID	Source	Used for
S18	Team's governance model PDF, 12 October 2026	Pipeline, flaws, pricing formula, standards table, viva answers
S19	Team's workflow description (this project)	Roles, workflow, evidence steps
S20	Dataful, analysis of CPCB data on e-waste processing by state (secondary)	States collecting less relative to size

Re-check every figure and legal point against the original source before quoting it. Laws, portals and prices change.